

Problem 1

Calculate the work function in electron volts of a metal, given that photoelectric threshold is (i) 6200 Å and (ii) 5000 Å.

Solution: Given

$$(i) \quad \lambda_0 = 6.2 \times 10^{-7} \text{ m}$$

$$(ii) \quad \lambda_0 = 5.0 \times 10^{-7} \text{ m}$$

$$(i) \quad \phi_0 = h\nu_0 = \frac{hc}{\lambda_0}$$

$$\begin{aligned} \phi_0 &= \frac{6.62 \times 10^{-34} \times 3 \times 10^8}{6.2 \times 10^{-7} \times 1.6 \times 10^{-19}} \text{ eV} \\ &= 2.0 \text{ eV} \end{aligned}$$

$$(ii) \quad \lambda_0 = 5.0 \times 10^{-7} \text{ m}$$

$$\phi_0 = \frac{6.62 \times 10^{-34} \times 3 \times 10^8}{5 \times 10^{-7} \times 1.6 \times 10^{-19}} \text{ eV} = 2.5 \text{ eV}$$

Calculate the wavelength associated with an electron subjected to a potential difference of 1.25 kV.

Solution: The de-Broglie wavelength λ is given by

$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2mE}}$$

$$m = 9.1 \times 10^{-31} \text{ kg}, \quad e^- = 1.6 \times 10^{-19} \text{ C}$$

$$\text{eV} = E = 1.25 \text{ K electron volts}$$

$$\lambda = \frac{6.62 \times 10^{-34}}{\sqrt{(2 \times 9.1 \times 10^{-31} \times 1.25 \times 10^3 \times 1.6 \times 10^{-19})}}$$

$$\lambda = 1.10 \times 10^{-9} \text{ m}$$

$$\lambda = 11 \text{ Å}$$

A spectral line has wavelength $4000 \text{ \AA}$. Calculate frequency and the energy in eV of the photon associated with it.

Solution:

$$\lambda = 4000 \times 10^{-10} \text{ m}$$

$$c = 3 \times 10^8 \text{ m/s}$$

$$h = 6.62 \times 10^{-34} \text{ J-s}$$

$$\begin{aligned}\nu &= \frac{c}{\lambda} \\ &= \frac{3 \times 10^8}{4000 \times 10^{-10}} \\ &= 0.75 \times 10^{15} \text{ Hz}\end{aligned}$$

The de-Broglie wavelength of the electron is $30 \text{ \AA}$. Calculate the phase velocity and group velocity of the matter wave associated with the electron.

Solution: For matter waves,

$$\lambda = \frac{h}{mv} \text{ or } v = \frac{h}{m\lambda}$$

$$\begin{aligned}v &= \frac{6.62 \times 10^{-34} \text{ J-s}}{(30 \times 10^{-10} \text{ m}) \times (9.1 \times 10^{-31} \text{ kg})} \\ &= 2.42 \times 10^5 \text{ m/s}\end{aligned}$$

Phase velocity,

$$v_p = \frac{v}{2} = 1.21 \times 10^5 \text{ m/s}$$

And,

$$\text{Group velocity } v_g = v = 2.42 \times 10^5 \text{ m/s.}$$

EXAMPLE 12.5

Calculate the Compton shift if X-rays of wavelength $1.0 \text{ \AA}$ are scattered from a carbon block. The scattered radiation is viewed at 90° to the incident beam.

Solution: Given $\lambda = 1.0 \text{ \AA} = 10^{-10} \text{ m}$ and $\phi = 90^\circ$

$$\begin{aligned}\Delta\lambda &= \frac{h}{m_0 c} (1 - \cos \phi) \\ &= \frac{6.62 \times 10^{-34}}{9.1 \times 10^{-31} \times 3 \times 10^8} (1 - \cos 90^\circ) \\ &= 0.0242 \text{ \AA}\end{aligned}$$

EXAMPLE 12.6

X-rays with $\lambda = 1 \text{ \AA}$ are scattered from a carbon block. The scattered radiation is viewed at 90° to the incident beam. What kinetic energy is imparted to the recoil electron?

Solution: Given $\lambda = 1 \times 10^{-10} \text{ m}$

We know that

$$\begin{aligned}\frac{hc}{\lambda} &= \frac{hc}{\lambda'} + E_k \\ \Rightarrow \frac{hc}{\lambda} &= \frac{hc}{\lambda + \Delta\lambda} + E_k \\ &= \frac{6.62 \times 10^{-37} \times 3 \times 10^8 \times 2.425 \times 10^{-12}}{1 \times 10^{-10} \times (1 + 0.02425) \times 10^{-10}} \\ &= 47.02 \times 10^{-18} \text{ J} \\ &= 294 \text{ eV}\end{aligned}$$

EXAMPLE 12.7

The average period that elapses between the excitation of an atom and the time it emits radiation is 10^{-8} sec. Find the uncertainty in energy emitted and the uncertainty in the frequency of light emitted.

Solution:

$$h = 6.63 \times 10^{-34} \text{ J-s}$$

The uncertainty in the energy is given by

$$\begin{aligned}\Delta E &= \frac{\hbar}{\Delta t} \\ &= \frac{6.63 \times 10^{-34}}{2\pi \times 10^{-8}} \\ &= 1.054 \times 10^{-26} \text{ J}\end{aligned}$$

The uncertainty in the frequency of light emitted is given by

$$\begin{aligned}\Delta \nu &= \frac{\Delta E}{h} = \frac{1.054 \times 10^{-26}}{6.63 \times 10^{-34}} \\ &= 1.59 \times 10^7 \text{ Hz}\end{aligned}$$

EXAMPLE 12.8

A proton is moving with a speed of 2×10^8 m/sec. Find the wavelength of matter wave associated with it.

Solution: Given

$$v = 2 \times 10^8 \text{ m/sec}$$

And

$$\begin{aligned}\lambda &= \frac{h}{mv} \\ &= \frac{6.62 \times 10^{-34}}{1.67 \times 10^{-27} \times 2 \times 10^8} \\ &= 1.98 \times 10^{-15} \text{ m}\end{aligned}$$

EXAMPLE 12.9

Prove uncertainty relation for energy and time from position-momentum uncertainty relation.

Solution: We know that

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

Consider a free particle of mass ' m ' moving with velocity v .

Its kinetic energy is,
$$E = \frac{1}{2}mv^2 = \frac{p^2}{2m}$$

Uncertainty in energy is given by

$$\Delta E = \frac{2p\Delta p}{2m}$$

$$\Delta E = \frac{p\Delta p}{m}$$

$$\Delta E = \frac{mv\Delta p}{m} = v\Delta p$$

But

$$v = \frac{\Delta x}{\Delta t}$$

$\Rightarrow$

$$\Delta E = \frac{\Delta x}{\Delta t} \Delta p$$

$\Rightarrow$

$$\Delta E \Delta t = \Delta x \Delta p$$

But

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

So,

$$\Delta E \Delta t \geq \frac{\hbar}{2}$$

SHORT ANSWER QUESTIONS

1. What is Planck's hypothesis?
2. Explain de-Broglie hypothesis.
3. What is the difference between phase velocity and group velocity?
4. What is de-Broglie wavelength?
5. What is Heisenberg uncertainty principle?
6. What do you understand by uncertainty principle?
7. Write a short note on F-D statistics.
8. Explain in brief B-E statistics.
9. What are Bosons and Fermions?
10. What do you mean by Compton effect?
11. What are quarks, gluons and hadrons?
12. What do you mean by Quantum Statistics?
13. Write a short note on photoelectric effect.
14. Name the experiments which show wave nature of light.
15. Explain the following terms: position space, momentum space and phase space

LONG ANSWER QUESTIONS

1. What is Planck's quantum hypothesis to explain the observed spectrum of a black body? Give the shortcomings of old quantum theory.
2. What are the shortcomings of classical theory? Derive formula for the de-Broglie wavelength of a particle.
3. What is the difference between phase and group velocities? Show that the de-Broglie group velocity associated with the wave packet is equal to the velocity of the particle.
4. Compare Bose-Einstein and Fermi-Dirac statistics.
5. Show that for a non-relativistic particle, the phase velocity is half of the group velocity.
6. Discuss and derive de-Broglie hypothesis. How it has been verified experimentally?
7. What are matter waves? Mention any three properties of matter waves. Describe the experiment of Davisson and Germer to demonstrate the wave nature of the electrons.
8. Derive an expression for group velocity with which a wave group travels.
9. What is Planck's constant?
10. Show that the group velocity and phase velocity are same in non-dispersive medium.
11. Describe G. P. Thomson experiment to prove the validity of the uncertainty principle and to show the wave nature of the electrons.
12. Explain Heisenberg's uncertainty principle and its important applications.
13. How do the group velocity and wave velocity differ? Show that the phase

velocity can exceed the velocity of light in vacuum.

14. Why can electron not exist in the nucleus? Why Heisenberg's principle is important for microscopic bodies alone?
15. State time energy uncertainty principle. Obtain its expression from position-momentum uncertainty principle.
16. Show that the group velocity of the particle is equal to the velocity of the particle. Derive the relation between ν_g and ν_p .

NUMERICAL PROBLEMS

1. Calculate the smallest possible uncertainty in the position of an electron moving with velocity 3×10^7 m/s.
Ans. 0.038 Å
2. The average life time of hydrogen in excited state is 2.5×10^{-14} s. Calculate the uncertainty in the measurement of energy in this state.
Ans. 0.025 eV
3. Calculate the de-Broglie wavelengths of a 2 eV photon and an electron with kinetic energy of 2 eV.
Ans. 8.6 Å
4. What will be the energy of a gamma ray photon having wavelength 1 Å?
Ans. 1.24×10^4 eV
5. A proton and a deuteron have the same kinetic energy. Which has a longer wavelength?
Ans. Proton
6. Find the phase and group velocities of an electron whose de-Broglie wavelength is 1.2 Å.
Ans. 3.03×10^6 m/sec and 6.06×10^6 m/sec
7. A certain spectral line has wavelength 4000 Å. Calculate the energy of the photon.
Ans. 4.9×10^{-19} J
8. The threshold frequency for photoelectric emission in copper is 1.1×10^{15} Hz. Find the maximum energy in eV when $\gamma = 1.2 \times 10^{15}$ Hz (frequency).
Ans. 0.414 eV
9. Show that the de-Broglie wave velocity is a function of wavelength in free space.
10. An electron has de-Broglie wavelength 2.0×10^{-12} m. Find its kinetic energy. Also find the phase and group velocity of de-Broglie waves.